KCE PATHWAYS · GRADE 3 · UNIT 6: BUILDING TOGETHER · WEEK 27 · MATH

Designing a Playground in Chattanooga — Teacher Lesson Plan & Answer Key

Learning objectives: Students find rectangles with the same perimeter but different areas and with the same area but different perimeters (the shape closest to a square has the greatest area for a given perimeter and the least perimeter for a given area), find the area of combined rectangles with the distributive property (7 × 13 = 7 × 8 + 7 × 5), find missing sides, the perimeter and the area of L-shaped rectilinear figures, and solve two-step area and perimeter problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce the new community playground near the Tennessee River in Chattanooga, Tennessee, with designer Mr. Conrad and his young planners Chloe, Chidi, Colton and Cora.
2. Warm-Up: Week 26 Review (10 pts): multiplying by multiples of 10 and the properties. If a student misses both, review “use the fact, then think in tens: 8 × 70 = 56 tens = 560.”
3. Picture It: Designing a Playground in Chattanooga. A fence or border goes around the edge (perimeter); mats, sand and tiles cover the inside (area). The Walnut Street Bridge, built in 1890, reopened in 1993 as a bridge for people on foot. Ask: is a fence around the edge or covering the inside?
4. Word Wrangler: 6 vocab cards (perimeter, area, square foot, dimensions, rectilinear figure, distributive property).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W27.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Same Perimeter, Different Areas (10 pts): with 24 feet of border, the length plus the width is 12, and the areas run from 11 to 36 square feet. Tip: list the rectangles in order so none is skipped; the one closest to a square holds the most.
7. Same Area, Different Perimeters (10 pts): use factor pairs (24 square feet: 1 by 24, 2 by 12, 3 by 8, 4 by 6). A long, skinny shape needs the most border; the one closest to a square needs the least.
8. Combined Rectangles and the Distributive Property (10 pts): split a rectangle into two parts, find each area and add (7 × 8 + 7 × 5 = 91), or subtract a missing corner from a big rectangle. Watch for multiplying only one part.
9. Design Challenge: L-Shaped Figures (10 pts): find a missing side by comparing opposite directions (6 + 4 = 10; 3 + 5 = 8), add all six sides for the perimeter, and split or subtract for the area. Some items use the L-shaped drawing on screen.
10. Planner Challenge: Area and Perimeter (10 pts): bags of mulch, fence cost, tiles around a drain, and a square stage. Ask first: is this job around the edge or covering the inside?
11. Bonus Practice: The Picnic Patio (XP only, not scored).

12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches area and perimeter vocabulary to meanings, Set 2 matches rectangles to their perimeters, Set 3 matches rectangles to their areas.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places and the title Mr., end mark, list commas, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: Building Day (10 pts, 5 each): finding an area from a perimeter, an L-shaped garden that is not split for you, the greatest area for 22 feet of rope, and which-explanation or find-the-mistake items (any split of 14 works; an L shape has six sides).
16. Boss Battle, Part 2: Ribbon Cutting (10 pts, 5 each): a missing length, a square's perimeter from its area, comparing borders, and find-the-mistake items where the student types the correct answer (a square with 6-foot sides has a perimeter of 24 feet, not 36).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W27.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Design Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Playground Hero!" with the congratulations message "Playground hero! You measured every part of the new playground!" and a big confetti burst; below 85 shows "Great Effort Today!" with encouragement ("Every designer gets better with practice."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 26 REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Each new swing seat costs $40. The playground buys 6 swing seats. How many dollars is that?	**240** (also accepted: $240, 240 dollars, $240.00, $ 240)	6 × 40 = 6 × 4 tens = 24 tens = 240. So it costs $240.
w2_s0	Set 1: Which property does this show? 3 × (2 × 50) = (3 × 2) × 50 [The associative property / The commutative property / The distributive property / It is not true, because the grouping changes the product.]	**The associative property**	The same three factors are grouped in a different way, and the product stays 300. That is the associative property.
w1_s1	Set 2: The path will have 8 rows of bricks with 70 bricks in each row. How many bricks is that?	**560** (also accepted: 560 bricks)	8 × 70 = 8 × 7 tens = 56 tens = 560 bricks.
w2_s1	Set 2: Find the mistake. Chidi says 5 × 40 = 20. What went wrong? [5 × 4 tens = 20 tens, and 20 tens is 200. He forgot that he was multiplying tens. / Nothing is wrong. 5 × 40 = 20. / 5 × 40 should be 45, because 5 + 40 = 45. / 5 × 40 should be 2,000, because you add two zeros.]	**5 × 4 tens = 20 tens, and 20 tens is 200. He forgot that he was multiplying tens.**	40 is 4 tens. 5 × 4 tens = 20 tens, and 20 tens = 200.
w1_s2	Set 3: Use the break-apart trick: 7 × 16 = 7 × 10 + 7 × 6. What is 7 × 16?	**112**	7 × 10 = 70 and 7 × 6 = 42. 70 + 42 = 112.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: Which is equal to 9 × 30? [30 × 9 / 9 + 30 / 9 × 3 / 93]	**30 × 9**	By the commutative property, the factors can switch places: 9 × 30 = 30 × 9 = 270.

SAME PERIMETER, DIFFERENT AREAS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p1_s0	Set 1: Cora and Colton each use 32 feet of border. Cora makes a 3 by 13 rectangle. Colton makes an 8 by 8 square. How many more square feet does Colton's square cover?	**25** (also accepted: 25 square feet, 25 sq ft, 25 sq. ft., 25 square foot, 25 square ft)	Both use 32 feet: 3 + 13 + 3 + 13 = 32 and 8 + 8 + 8 + 8 = 32. The areas are 3 × 13 = 39 and 8 × 8 = 64 square feet. 64 − 39 = 25 more square feet.
p2_s0	Set 1: Each rectangle below has a perimeter of 28 feet. Which one has the greatest area? [7 feet by 7 feet / 4 feet by 10 feet / 2 feet by 12 feet / 1 foot by 13 feet]	**7 feet by 7 feet**	7 × 7 = 49, 4 × 10 = 40, 2 × 12 = 24, and 1 × 13 = 13 square feet. The square holds the most.
p1_s1	Set 2: Two steps! A rectangular stage has a perimeter of 18 feet. It is 2 feet wide. What is its area, in square feet?	**14** (also accepted: 14 square feet, 14 sq ft, 14 sq. ft., 14 square foot, 14 square ft)	Step 1: The two widths are 2 + 2 = 4 feet. 18 − 4 = 14 feet for the two lengths, so each length is 7 feet. Step 2: Area = 2 × 7 = 14 square feet.
p2_s1	Set 2: How many different rectangles with whole-number sides have a perimeter of 12 feet? (A square counts as a rectangle.)	**3** (also accepted: 3 rectangles)	Half the perimeter is 6 feet, so the length plus the width is 6: 1 by 5, 2 by 4, and 3 by 3. That is 3 rectangles.
p1_s2	Set 3: A 3 by 12 rectangle and a 6 by 9 rectangle both have a perimeter of 30 feet. How many more square feet does the 6 by 9 rectangle cover?	**18** (also accepted: 18 square feet, 18 sq ft, 18 sq. ft., 18 square foot, 18 square ft)	6 × 9 = 54 and 3 × 12 = 36 square feet. 54 − 36 = 18 more square feet.
p2_s2	Set 3: Which pair of rectangles has the same perimeter? [2 by 8 and 4 by 6 / 2 by 8 and 4 by 4 / 3 by 5 and 5 by 5 / 1 by 9 and 3 by 3]	**2 by 8 and 4 by 6**	2 + 8 + 2 + 8 = 20 and 4 + 6 + 4 + 6 = 20. Both perimeters are 20 feet, but the areas are 16 and 24 square feet.

SAME AREA, DIFFERENT PERIMETERS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
q1_s0	Set 1: Two steps! The splash pad has an area of 36 square feet and is 4 feet wide. What is its perimeter, in feet?	**26** (also accepted: 26 feet, 26 ft, 26 ft.)	Step 1: 4 × 9 = 36, so it is 9 feet long. Step 2: 4 + 9 + 4 + 9 = 26 feet.
q2_s0	Set 1: Which explanation is correct? Colton wants a rectangular mat with whole-number sides that covers 36 square feet and needs the least border. [Choose 6 feet by 6 feet. The shape closest to a square needs the least border, only 24 feet. / Choose 1 foot by 36 feet, because long, thin mats need the least border. / Choose any of them, because every mat that covers 36 square feet needs the same border. / Choose 2 feet by 18 feet, because 2 + 18 = 20 is the smallest number.]	**Choose 6 feet by 6 feet. The shape closest to a square needs the least border, only 24 feet.**	The perimeters are 74 feet for 1 by 36, 40 feet for 2 by 18, 30 feet for 3 by 12, 26 feet for 4 by 9, and 24 feet for 6 by 6. The square needs the least border.

ID	Question	Answer	Why (explain to student)
q1_s1	Set 2: A 3 by 6 sandbox and a 2 by 9 sandbox both cover 18 square feet. What is the perimeter of the 2 by 9 sandbox, in feet?	**22** (also accepted: 22 feet, 22 ft, 22 ft.)	2 + 9 + 2 + 9 = 22 feet. (The 3 by 6 sandbox has a perimeter of only 18 feet.)
q2_s1	Set 2: Find the mistake. Chloe says every rectangle with an area of 12 square feet has a perimeter of 14 feet. What went wrong? [Only the 3 by 4 rectangle has a perimeter of 14 feet. The 2 by 6 has 16 feet and the 1 by 12 has 26 feet. / Nothing is wrong. Same area always means same perimeter. / They all have a perimeter of 12 feet, the same as the area. / They all have a perimeter of 24 feet, because 12 + 12 = 24.]	**Only the 3 by 4 rectangle has a perimeter of 14 feet. The 2 by 6 has 16 feet and the 1 by 12 has 26 feet.**	Same area does not mean same perimeter. 3 + 4 + 3 + 4 = 14, 2 + 6 + 2 + 6 = 16, and 1 + 12 + 1 + 12 = 26.
q1_s2	Set 3: Think of every rectangle with whole-number sides and an area of 20 square feet. What is the smallest perimeter one of them can have, in feet?	**18** (also accepted: 18 feet, 18 ft, 18 ft.)	The rectangles are 1 by 20 (42 feet), 2 by 10 (24 feet), and 4 by 5 (18 feet). The smallest perimeter is 18 feet.
q2_s2	Set 3: A 2 by 16 rectangle and a 4 by 8 rectangle both cover 32 square feet. How many more feet of border does the 2 by 16 rectangle need?	**12** (also accepted: 12 feet, 12 ft, 12 ft.)	2 + 16 + 2 + 16 = 36 feet and 4 + 8 + 4 + 8 = 24 feet. 36 − 24 = 12 more feet.

COMBINED RECTANGLES AND THE DISTRIBUTIVE PROPERTY (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: A rubber mat is made of two parts side by side: one part is 4 feet by 6 feet and the other is 4 feet by 3 feet. What is the total area, in square feet?	**36** (also accepted: 36 square feet, 36 sq ft, 36 sq. ft., 36 square foot, 36 square ft)	4 × 6 = 24 and 4 × 3 = 12. 24 + 12 = 36 square feet. That is the same as 4 × 9 = 36.
r2_s0	Set 1: A 5 by 12 rectangle is split into a 5 by 10 part and a 5 by 2 part. Which equation finds its area? [5 × 12 = 5 × 10 + 5 × 2 / 5 × 12 = 5 × 10 + 2 / 5 × 12 = 5 + 10 + 2 / 5 × 12 = 5 × 10 × 2]	**5 × 12 = 5 × 10 + 5 × 2**	Both parts are 5 feet tall, so multiply 5 by each part and add: 50 + 10 = 60 square feet.
r1_s1	Set 2: Two rubber mats are both 8 feet wide. One is 6 feet long and the other is 4 feet long. Chidi pushes them together end to end and says the total area is 8 × 10. What is the total area, in square feet?	**80** (also accepted: 80 square feet, 80 sq ft, 80 sq. ft., 80 square foot, 80 square ft)	8 × 6 + 8 × 4 = 8 × (6 + 4) = 8 × 10 = 80 square feet. Check: 48 + 32 = 80.
r2_s1	Set 2: Find the mistake. Colton finds the area of a 6 by 9 rectangle by splitting 9 into 7 + 2. He writes 6 × 7 + 2 = 44. What went wrong? [He forgot to multiply the 2 by 6. 6 × 7 + 6 × 2 = 42 + 12 = 54 square feet. / Nothing is wrong. The area is 44 square feet. / He should add the sides: 6 + 7 + 2 = 15. / He should find 6 × 7 × 2 = 84.]	**He forgot to multiply the 2 by 6. 6 × 7 + 6 × 2 = 42 + 12 = 54 square feet.**	Both smaller rectangles are 6 feet tall. 6 × 7 = 42 and 6 × 2 = 12. 42 + 12 = 54 = 6 × 9.
r1_s2	Set 3: A path is 9 feet wide and 14 feet long. Split 14 into 10 + 4. What is the area of the path, in square feet?	**126** (also accepted: 126 square feet, 126 sq ft, 126 sq. ft., 126 square foot, 126 square ft)	9 × 10 = 90 and 9 × 4 = 36. 90 + 36 = 126 square feet.

ID	Question	Answer	Why (explain to student)
r2_s2	Set 3: A garden starts as a 10 by 6 rectangle, but a 4 by 3 corner is cut out for a bench. What is the area of the garden that is left, in square feet?	**48** (also accepted: 48 square feet, 48 sq ft, 48 sq. ft., 48 square foot, 48 square ft)	Whole rectangle: 10 × 6 = 60. Corner: 4 × 3 = 12. 60 − 12 = 48 square feet.

DESIGN CHALLENGE: L-SHAPED FIGURES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Look at the L-shaped sandbox. What is its perimeter, in feet?	**26** (also accepted: 26 feet, 26 ft, 26 ft.)	Add all six sides: 4 + 2 + 3 + 4 + 7 + 6 = 26 feet.
s2_s0	Set 1: What is the area of the same L-shaped sandbox, in square feet? (Its sides are 4, 2, 3, 4, 7, and 6 feet.)	**36** (also accepted: 36 square feet, 36 sq ft, 36 sq. ft., 36 square foot, 36 square ft)	Big rectangle: 7 × 6 = 42. Missing corner: 3 × 2 = 6. 42 − 6 = 36 square feet. Or split it: 4 × 6 + 3 × 4 = 24 + 12 = 36.
s1_s1	Set 2: Mr. Conrad has exactly 30 feet of fence for a rectangular toddler play yard with whole-number sides. Which design uses all 30 feet and has the greatest area? [7 feet by 8 feet / 5 feet by 10 feet / 3 feet by 12 feet / 8 feet by 8 feet]	**7 feet by 8 feet**	7 by 8, 5 by 10, and 3 by 12 all use 30 feet, with areas of 56, 50, and 36 square feet. 8 by 8 needs 32 feet of fence, which is too much.
s2_s1	Set 2: An L-shaped bench area has a bottom of 9 feet and a left side of 7 feet. The top is 5 feet, the step down is 3 feet, and the step across is 4 feet. How long is the right side, in feet?	**4** (also accepted: 4 feet, 4 ft, 4 ft.)	The left side is 7 feet. The right side plus the step down must also make 7 feet: 7 − 3 = 4 feet.
s1_s2	Set 3: A rectangular garden must cover 30 square feet and have whole-number sides. The planners want to use the least border. What perimeter will it have, in feet?	**22** (also accepted: 22 feet, 22 ft, 22 ft.)	The rectangles are 1 by 30 (62 feet), 2 by 15 (34 feet), 3 by 10 (26 feet), and 5 by 6 (22 feet). The least border is 22 feet.
s2_s2	Set 3: Which explanation is correct? Cora cuts a small rectangle out of one corner of a rectangular mat. [The area gets smaller, but the perimeter stays the same, because the two new edges are just as long as the two edges that were cut away. / The area and the perimeter both get smaller. / The area stays the same, but the perimeter gets bigger. / Nothing changes, because it is still the same mat.]	**The area gets smaller, but the perimeter stays the same, because the two new edges are just as long as the two edges that were cut away.**	A 10 by 8 rectangle has a perimeter of 36 feet. Cut out a 4 by 3 corner and the sides are 6, 3, 4, 5, 10, and 8. That adds up to 36 feet too, but the area drops from 80 to 68 square feet.

PLANNER CHALLENGE: AREA AND PERIMETER (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! The swing area is 8 feet by 7 feet. One bag of rubber mulch covers 4 square feet. How many bags are needed?	**14** (also accepted: 14 bags)	Step 1: 8 × 7 = 56 square feet. Step 2: 56 ÷ 4 = 14 bags.
c2_s0	Set 1: Two steps! A fence will go all the way around a 9 by 6 garden. Each foot of fence costs $3. How many dollars will the fence cost?	**90** (also accepted: $90, 90 dollars, $90.00, $ 90)	Step 1: 9 + 6 + 9 + 6 = 30 feet of fence. Step 2: 30 × $3 = $90.
c1_s1	Set 2: The splash pad is 10 feet by 6 feet. A 2 by 3 drain area in one corner will not get tiles. How many square feet will get tiles?	**54** (also accepted: 54 square feet, 54 sq ft, 54 sq. ft., 54 square foot, 54 square ft)	10 × 6 = 60 and 2 × 3 = 6. 60 − 6 = 54 square feet.

ID	Question	Answer	Why (explain to student)
c2_s1	Set 2: Two steps! A square stage has a perimeter of 32 feet. What is its area, in square feet?	**64** (also accepted: 64 square feet, 64 sq ft, 64 sq. ft., 64 square foot, 64 square ft)	Step 1: A square has 4 equal sides, and 32 ÷ 4 = 8 feet. Step 2: 8 × 8 = 64 square feet.
c1_s2	Set 3: The garden is 7 feet wide and 12 feet long. Use the distributive property to find its area, in square feet.	**84** (also accepted: 84 square feet, 84 sq ft, 84 sq. ft., 84 square foot, 84 square ft)	7 × 12 = 7 × 10 + 7 × 2 = 70 + 14 = 84 square feet.
c2_s2	Set 3: Two steps! A rectangle covers 42 square feet and is 6 feet wide. What is its perimeter, in feet?	**26** (also accepted: 26 feet, 26 ft, 26 ft.)	Step 1: 6 × 7 = 42, so it is 7 feet long. Step 2: 6 + 7 + 6 + 7 = 26 feet.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match perimeter	**the distance around a shape**	2 points if matched on the first try.
match_s0_2	Set 1: Match area	**the square units that cover a shape**	2 points if matched on the first try.
match_s0_3	Set 1: Match square foot	**a square 1 foot long on each side**	2 points if matched on the first try.
match_s0_4	Set 1: Match dimensions	**the length and width of a rectangle**	2 points if matched on the first try.
match_s0_5	Set 1: Match rectilinear figure	**an L shape made of rectangles**	2 points if matched on the first try.
match_s1_1	Set 2: Match 3 by 9 rectangle	**perimeter 24 feet**	2 points if matched on the first try.
match_s1_2	Set 2: Match 5 by 5 square	**perimeter 20 feet**	2 points if matched on the first try.
match_s1_3	Set 2: Match 2 by 7 rectangle	**perimeter 18 feet**	2 points if matched on the first try.
match_s1_4	Set 2: Match 1 by 15 rectangle	**perimeter 32 feet**	2 points if matched on the first try.
match_s1_5	Set 2: Match 4 by 10 rectangle	**perimeter 28 feet**	2 points if matched on the first try.
match_s2_1	Set 3: Match 3 by 8 rectangle	**24 square feet**	2 points if matched on the first try.
match_s2_2	Set 3: Match 6 by 6 square	**36 square feet**	2 points if matched on the first try.
match_s2_3	Set 3: Match 5 by 7 rectangle	**35 square feet**	2 points if matched on the first try.
match_s2_4	Set 3: Match 2 by 9 rectangle	**18 square feet**	2 points if matched on the first try.
match_s2_5	Set 3: Match 4 by 4 square	**16 square feet**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: chattanooga is on the tennessee river	**Chattanooga is on the Tennessee River.**	Chattanooga is the name of a city and Tennessee River is the name of a river, so they get capital letters. End the telling sentence with a period.

ID	Question	Answer	Why (explain to student)
fs2_s0	Set 1: Fix the sentence: the playground will have swings slides and a sandbox	**The playground will have swings, slides, and a sandbox.** (also accepted: The playground will have swings, slides and a sandbox.)	Start with a capital T, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how long is the fence	**How long is the fence?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: chloe chidi and cora measured the sandbox	**Chloe, Chidi, and Cora measured the sandbox.** (also accepted: Chloe, Chidi and Cora measured the sandbox.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: did mr. conrad walk across the walnut street bridge	**Did Mr. Conrad walk across the Walnut Street Bridge?**	Start with a capital D. Mr. Conrad is a name, and Walnut Street Bridge is the name of a bridge, so they get capital letters. The sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: the swings is next to the slide	**The swings are next to the slide.**	Start with a capital T and end with a period. Swings means more than one, so use are, not is.

BOSS BATTLE: BUILDING DAY (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! A rectangular reading nook has a perimeter of 26 feet and is 5 feet wide. What is its area, in square feet?	**40** (also accepted: 40 square feet, 40 sq ft, 40 sq. ft., 40 square foot, 40 square ft)	Step 1: 26 − 5 − 5 = 16 feet for the two lengths, so each length is 8 feet. Step 2: 5 × 8 = 40 square feet.
bb2_s0	Set 1: Which explanation is correct? Chidi and Cora find the area of a 6 by 14 deck. Chidi writes 6 × 10 + 6 × 4. Cora writes 6 × 7 + 6 × 7. [Both are right. 60 + 24 = 84 and 42 + 42 = 84, because 10 + 4 and 7 + 7 both make 14. / Only Chidi is right, because you must always split off a 10. / Only Cora is right, because the two parts must be the same size. / Neither is right, because 6 × 14 = 6 × 10 + 4 = 64.]	**Both are right. 60 + 24 = 84 and 42 + 42 = 84, because 10 + 4 and 7 + 7 both make 14.**	You can split 14 into any two parts and multiply each part by 6. 6 × 10 + 6 × 4 = 84 and 6 × 7 + 6 × 7 = 84. Both equal 6 × 14 = 84 square feet.
bb1_s1	Set 2: Two steps! Look at the L-shaped garden. It is not split for you. What is its area, in square feet?	**57** (also accepted: 57 square feet, 57 sq ft, 57 sq. ft., 57 square foot, 57 square ft)	Split it with a line going down: a 6 by 7 rectangle, 6 × 7 = 42, and a 3 by 5 rectangle, 3 × 5 = 15. 42 + 15 = 57 square feet. Or subtract: 9 × 7 = 63, and the missing corner is 3 × 2 = 6. 63 − 6 = 57.
bb2_s1	Set 2: Cora writes 7 × 15 = 7 × 10 + 7 × 5 to find the area of a 7 by 15 path. What is the area, in square feet?	**105** (also accepted: 105 square feet, 105 sq ft, 105 sq. ft., 105 square foot, 105 square ft)	7 × 10 = 70 and 7 × 5 = 35. 70 + 35 = 105 square feet.

ID	Question	Answer	Why (explain to student)
bb1_s2	Set 3: Colton has 22 feet of rope to outline a rectangle with whole-number sides. What is the greatest area he can outline, in square feet?	**30** (also accepted: 30 square feet, 30 sq ft, 30 sq. ft., 30 square foot, 30 square ft)	Length + width = 11: 1 by 10 (10), 2 by 9 (18), 3 by 8 (24), 4 by 7 (28), and 5 by 6 (30). The greatest area is 30 square feet.
bb2_s2	Set 3: Find the mistake. Chidi finds the perimeter of this L-shaped sandbox. He adds 5 + 4 + 8 + 6 = 23 feet. What went wrong? [He left out the 2-foot and 3-foot sides of the step. All six sides: 5 + 2 + 3 + 4 + 8 + 6 = 28 feet. / Nothing is wrong. The perimeter is 23 feet. / He should multiply: 8 × 6 = 48 feet. / He should add only the two longest sides: 8 + 6 = 14 feet.]	**He left out the 2-foot and 3-foot sides of the step. All six sides: 5 + 2 + 3 + 4 + 8 + 6 = 28 feet.**	An L shape has six sides, and the perimeter is the whole distance around. 5 + 2 + 3 + 4 + 8 + 6 = 28 feet.

BOSS BATTLE: RIBBON CUTTING (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: A rectangular sandbox covers 63 square feet and is 7 feet wide. How long is it, in feet?	**9** (also accepted: 9 feet, 9 ft, 9 ft.)	7 × 9 = 63, so the sandbox is 9 feet long.
bb4_s0	Set 1: Find the mistake. Chloe says a square with 6-foot sides has a perimeter of 36 feet. Type the correct perimeter, in feet.	**24** (also accepted: 24 feet, 24 ft, 24 ft.)	She found the area, 6 × 6 = 36 square feet. The perimeter is the distance around: 6 + 6 + 6 + 6 = 24 feet.
bb3_s1	Set 2: Two steps! A square sandbox covers 49 square feet. What is its perimeter, in feet?	**28** (also accepted: 28 feet, 28 ft, 28 ft.)	Step 1: 7 × 7 = 49, so each side is 7 feet. Step 2: 4 × 7 = 28 feet.
bb4_s1	Set 2: Find the mistake. Colton says a 10 by 6 rectangle with a 2 by 2 corner cut out covers 64 square feet. Type the correct area, in square feet.	**56** (also accepted: 56 square feet, 56 sq ft, 56 sq. ft., 56 square foot, 56 square ft)	He added the corner instead of subtracting it. 10 × 6 = 60, and 60 − 4 = 56 square feet.
bb3_s2	Set 3: A 4 by 9 rectangle and a 6 by 6 square both cover 36 square feet. How many more feet of border does the 4 by 9 rectangle need?	**2** (also accepted: 2 feet, 2 ft, 2 ft.)	4 + 9 + 4 + 9 = 26 feet and 6 + 6 + 6 + 6 = 24 feet. 26 − 24 = 2 more feet.
bb4_s2	Set 3: Find the mistake. Cora says the area of an 8 by 12 deck is 8 × 10 + 2 = 82 square feet. Type the correct area, in square feet.	**96** (also accepted: 96 square feet, 96 sq ft, 96 sq. ft., 96 square foot, 96 square ft)	Multiply both parts by 8: 8 × 10 = 80 and 8 × 2 = 16. 80 + 16 = 96 square feet.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W27.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 4 benches for $60 each; show how you used tens	$240. 4 × 6 = 24, so 4 × 6 tens = 24 tens = 240.
2	Sandbox 5 feet long and 3 feet wide: perimeter	16 feet (5 + 3 + 5 + 3 = 16).
3	Area of the same 5 by 3 sandbox	15 square feet (5 × 3 = 15).

#	Prompt	Model answer / grading note
4	A fence goes around the swings: area, perimeter or weight?	perimeter
5	Draw two different rectangles that each cover 12 square units	Any two of 1 by 12, 2 by 6 and 3 by 4, with the length and width labeled. Accept any reasonable attempt before the lesson.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Every rectangle with whole-number sides and a perimeter of 26 feet	Length + width = 13: 1 by 12 (12), 2 by 11 (22), 3 by 10 (30), 4 by 9 (36), 5 by 8 (40) and 6 by 7 (42 square feet). The 6 by 7 rectangle has the greatest area.
2	2 by 9 and 3 by 6 sandboxes (18 square feet each): which needs less border?	The 3 by 6 sandbox. Its perimeter is 18 feet and the 2 by 9 sandbox's is 22 feet, so it needs 4 feet less.
3	Colton says the area of a 7 by 12 garden is 7 × 10 + 2 = 72	He forgot to multiply the 2 by 7. 7 × 10 + 7 × 2 = 70 + 14 = 84 square feet.
4	L shape made of a 6 by 4 rectangle and a 2 by 3 rectangle	30 square feet. 6 × 4 = 24 and 2 × 3 = 6, and 24 + 6 = 30. Check that the drawing joins the two rectangles into an L.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Exactly 20 feet of border: the rectangle with the greatest area	5 feet by 5 feet, 25 square feet. Length + width = 10, and the areas are 9, 16, 21, 24 and 25 square feet; the shape closest to a square (here a square) holds the most.